Graphene nanoribbons

We consider the polarizability of an isolated graphene nanoribbon as described here Adaptive multi-spectral mimicking with 2D-material nanoresonator networks (https://iopscience.iop.org/article/10.1088/2040-8986/ad4722/pdf)

$$\alpha(\omega) = 2W^3 \frac{A_1}{\zeta_1 - \zeta(\omega)}, \text{ with } \zeta(\omega) = \frac{2i\varepsilon_0\varepsilon_B\omega W}{\sigma(\omega)}$$

Note that $\sigma(\omega)$ has units of $c\varepsilon_0$ and $\omega W\varepsilon_0$ clearly also has units of $c\varepsilon_0$, which means that $\zeta(\omega)$ is unitless, as required. Furthermore, we see quite clearly that $\alpha(\omega)$ has a resonance at the plasmon frequency, ω_p, which is given by the implicit relation:

$$\sigma(\omega)/\omega = 2i\varepsilon_0\varepsilon_B W/\zeta_1 \rightarrow \frac{e^2 E_F}{\pi\hbar^2\omega(\omega + i/\tau)} = \frac{2\varepsilon_0\varepsilon_B W}{\zeta_1}$$

If we take $W = 0.33\mu m$, $E_F = 0.5eV$ and $\varepsilon_B = 1$ (and neglect scattering), we obtain:

$$\omega^2 = \frac{e^2 E_F \zeta_1}{2\pi\hbar^2 W\varepsilon_0} = \frac{2\alpha c E_F \zeta}{W\hbar} \approx \frac{(2/137)(0.5\text{eV})(2.32)(3 \times 10^{14}\mu\text{m/s})}{(0.33\mu\text{m})(6.7 \times 10^{-16}\text{eV} \times \text{s})} \approx 0.022 \times 10^{30}/\text{s}^2$$

From which we get $\omega \approx 0.15 \times 10^{15}\text{s}^{-1}$ and $f \equiv \omega/(2\pi) \approx 24\text{THz}$ in agreement with Tunable mid-infrared coherent perfect absorption in a graphene meta-surface (https://www.nature.com/articles/srep13956.pdf).

Radiation term in the polarizability

We first solve **Problem 8.5** in the book by **Novotny and Hecht**, using their notation below. The "self force" and "self field are given as follows:

$$\mathbf{F}_{\text{self}} = \frac{q^2\dddot{\mathbf{r}}}{6\pi\varepsilon_0 c^3} = \frac{q\dddot{\mathbf{p}}}{6\pi\varepsilon_0 c^3} \rightarrow \mathbf{E}_{\text{self}} = \frac{\dddot{\mathbf{p}}}{6\pi\varepsilon_0 c^3}$$

If we have $\mathbf{p} = \alpha(\mathbf{E}_0 + \mathbf{E}_{\text{self}}) = \alpha_{\text{eff}}(\mathbf{E}_0)$, which defines the polarizability and the effeective polarizability, using $\dddot{\mathbf{p}} = i\omega^3\mathbf{p}$, we obtain:

$$\mathbf{p} = \alpha(\mathbf{E}_0 + i\omega^3\mathbf{p}/6\pi\varepsilon_0 c^3) \rightarrow \mathbf{p}\left[1 - \frac{i\alpha(\omega)\omega^3}{6\pi\varepsilon_0 c^3}\right] = \alpha(\omega)\mathbf{E}_0 \rightarrow$$

$$\alpha_{\text{eff}}(\omega) = \frac{\alpha(\omega)}{1 - i\alpha(\omega)k^3/6\pi\varepsilon_0},$$

which is the desired result given by **Novotny and Hecht**.

Optical Theorem

In this section, we prove the optical theorem, since none of the papers/textbooks I've read seem to do a good job of it. We begin with the extinction power, which is the surface integral of the cross term, in the Poynting vector, of the incident and scattered fields.

$$\frac{1}{2}\int \Re\Bigg[\mathbf{E}_i(\mathbf{r})\times\mathbf{H}_s^*(\mathbf{r})+\mathbf{E}_s(\mathbf{r})\times\mathbf{H}_i^*(\mathbf{r})\Bigg]\cdot\mathbf{n}\mathrm{d}a =$$

$$\frac{1}{2}\Re\int\nabla\cdot\Bigg[\mathbf{E}_i(\mathbf{r})\times\mathbf{H}_s^*(\mathbf{r})+\mathbf{E}_s(\mathbf{r})\times\mathbf{H}_i^*(\mathbf{r})\Bigg]\mathrm{d}\Omega =$$

$$\frac{1}{2}\Re\int\Big[\mathbf{H}_s^*(\mathbf{r})\cdot\nabla\times\mathbf{E}_i(\mathbf{r})-\mathbf{E}_i(\mathbf{r})\cdot\nabla\times\mathbf{H}_s^*(\mathbf{r})+\mathbf{H}_i^*(\mathbf{r})\cdot\nabla\times\mathbf{E}_s(\mathbf{r})-\mathbf{E}_s(\mathbf{r})\cdot\nabla\times\mathbf{H}_i^*(\mathbf{r})\Big]\mathrm{d}\Omega,$$

where, in the second line, we used the divergence theorem and in the third line we used the identity $\nabla\cdot(\mathbf{A}\times\mathbf{B})=\mathbf{B}\cdot(\nabla\times\mathbf{A})-\mathbf{A}\cdot(\nabla\times\mathbf{B})$. At this point, we may use the fact that $\nabla\times\mathbf{E}_i(\mathbf{r})=i\omega\mu_0\mathbf{H}_i(\mathbf{r})$, $\nabla\times\mathbf{E}_s(\mathbf{r})=i\omega\mu_0\mathbf{H}_s(\mathbf{r})$, $\nabla\times\mathbf{H}_s^*(\mathbf{r})=\mathbf{J}^*(\mathbf{r})+i\omega\varepsilon_0\mathbf{E}_s^*(\mathbf{r})$ and $\nabla\times\mathbf{H}_i^*(\mathbf{r})=i\omega\varepsilon_0\mathbf{E}_i(\mathbf{r})$. Thus, we obtain:

$$\frac{1}{2}\Re\int\Big[i\omega\mu_0(\mathbf{H}_s^*(\mathbf{r})\cdot\mathbf{H}_i(\mathbf{r})+\mathbf{H}_i^*(\mathbf{r})\cdot\mathbf{H}_s(\mathbf{r}))-i\omega\varepsilon_0(\mathbf{E}_i(\mathbf{r})\cdot\mathbf{E}_s^*(\mathbf{r})+\mathbf{E}_s(\mathbf{r})\cdot\mathbf{E}_i^*(\mathbf{r}))$$
$$-\mathbf{E}_i(\mathbf{r})\cdot\mathbf{J}^*(\mathbf{r})\Big]\mathrm{d}\Omega=-\frac{1}{2}\Re\int\mathbf{E}_i(\mathbf{r})\cdot\mathbf{J}^*(\mathbf{r})d\Omega$$

From **Equation 3.19 and 3.20** of **Bohren and Huffman**, we obtain:

$$\frac{1}{2}\Re\int\mathbf{E}_i(\mathbf{r})\cdot\mathbf{J}^*(\mathbf{r})\mathrm{d}\Omega=\frac{1}{2}\int\Re\Bigg[\mathbf{E}_s(\mathbf{r})\times\mathbf{H}_s^*(\mathbf{r})\Bigg]\cdot\mathbf{n}\mathrm{d}a$$

This is **Equation 2** of the paper Radiative corrections to the polarizability tensor of an electrically small anisotropic dielectric particle (https://opg.optica.org/directpdfaccess/3b008e82-9061-4bcb-97aba5de6456a2e6_195394/oe-18-4-3556.pdf?da=1&id=195394&seq=0&mobile=no) by **S. Albaladejo, R. Gomez-Medina, et al** (2010). Writing the above in terms of the polarizability, we obtain the following:

$$\frac{1}{2}\Re\int\mathbf{E}_i(\mathbf{r})\cdot\mathbf{J}^*(\mathbf{r})\mathrm{d}\Omega=\frac{1}{2}\varepsilon_0\Re(i\omega\alpha^*(\omega)|\mathbf{E}(\mathbf{r}_0)|^2)=\frac{1}{2}\varepsilon_0\omega\Im\alpha(\omega)|\mathbf{E}(\mathbf{r}_0)|^2$$

We also have:

$$\mathbf{H}_s(\mathbf{r})=\frac{\omega\varepsilon_0\alpha(\omega)E(\mathbf{r}_0)}{4}\nabla\times\Bigg[H_0^{(1)}(\omega\rho/c)[\cos(\theta)\mathbf{e}_\rho-\sin(\theta)\mathbf{e}_\theta]\Bigg]=$$

$$\frac{\omega\varepsilon_0\alpha(\omega)\sin(\theta)E(\mathbf{r}_0)}{4\rho}\Bigg[-\partial_\rho(\rho H_0^{(1)}(\omega\rho/c))+H_0(\rho\omega/c)\Bigg]\mathbf{e}_z=$$

$$\frac{\omega^2\varepsilon_0\alpha(\omega)\sin(\theta)E(\mathbf{r}_0)}{4c}H_1^{(1)}(\rho\omega/c)\mathbf{e}_z,$$

$$\mathbf{E}_s(\mathbf{r})=\frac{1}{\omega}\frac{\omega^3\alpha(\omega)\sin(\theta)E(\mathbf{r}_0)}{4c^2}H_1^{(1)}(\rho\omega/c)\mathbf{e}_\theta$$

Note that in the above (and the below), we use asymptotic forms for the cylindrical Hankel functions. By integrating the scattered field Poynting vector, we readily obtain:

$$\frac{\omega^3|\alpha(\omega)|^2\varepsilon_0}{16c^2} = \frac{1}{2}\varepsilon_0\omega\Im\alpha(\omega) \rightarrow \frac{\omega^2|\alpha(\omega)|^2}{8c^2} = \Im\alpha(\omega) \rightarrow \Im\frac{1}{\alpha(\omega)} = -\frac{\omega^2}{8c^2},$$

where the last equality follows from $\Im(1/x) = -\Im(x)/|x|^2$.

Let us now revisit the nanoribbon polarizability:

$$\alpha(\omega) = 2W^3\frac{A_1}{\zeta_1 - \zeta(\omega)},\ \text{with}\ \zeta(\omega) = \frac{2i\varepsilon_0\varepsilon_B\omega W}{\sigma(\omega)}$$

To satisfy the optical theorem, we see that the polarizability needs to be amended to give:

$$\alpha(\omega) = 2W^3\frac{A_1}{\zeta_1 - \zeta(\omega) - iA_1W^2\omega^2/(4c^2)}$$

Consistency with requirement of no material loss

Requiring no loss in the material gives us the following condition:

$$\Re\int \mathbf{E}_i(\mathbf{r})\cdot\mathbf{J}^*(\mathbf{r})\mathrm{d}\Omega = -\Re\int \mathbf{E}_s(\mathbf{r})\cdot\mathbf{J}^*(\mathbf{r})\mathrm{d}\Omega \rightarrow$$

$$\Re\Big[i\omega\alpha^*(\omega)\Big]|\mathbf{E}_i(0)|^2 = \Re\Big[-i\omega\alpha^*(\omega)\mathbf{E}_s(0)\cdot\mathbf{E}_i^*(0)\Big]$$

Taking the external electric field to have p-polarization (i.e. $\mathbf{E}_i(\mathbf{r}) \equiv E_i(\mathbf{r})\mathbf{e}_x$), we obtain the following:

$$\mathbf{E}_s(0)\cdot\mathbf{E}_i^*(0) = \alpha(\omega)|\mathbf{E}_i(0)|^2\left[\frac{\omega^2}{c^2} + \partial_x\partial_x\right]\frac{i}{4}H_0(\rho\omega/c)$$

Thus we obtain the following:

$$\Im\alpha(\omega) = \frac{|\alpha(\omega)|^2}{4}\Re\left[\left[\frac{\omega^2}{c^2} + \partial_x\partial_x\right]J_0(\rho\omega/c)\right]$$

Using $\partial_x J_0(\rho\omega/c) = -\frac{\omega}{c}\frac{x}{\rho}J_1(\rho\omega/c)) \approx -\frac{\omega^2}{2c^2}x$, we obtain:

$$\Im\alpha(\omega) = \frac{\omega^2|\alpha(\omega)|^2}{8c^2},$$

whichs is precisely the optical theorem.